

An Intelligent Misinformation Detection System Using Machine Learning and Natural Language Processing

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Abstract

Social media and the internet are everywhere these days, people use them all the time. Most folks just grab info from online without checking if its real or not. That leads to so much fake or wrong stuff floating around out there. It messes things up for everyone in the world, kind of a big problem. I think techniques to spot this misinformation are super important right now. This thesis is going to focus on using machine learning algorithms to detect it. Well review different ones, looking at how they work with labeled data and unlabeled data too. The goal is to see if these algorithms actually apply well in practice. Different ways to measure the features will come into play. And measures like accuracy, precision, recall, and the F1 score stats, those will be key parts of the whole thing. It seems like without verifying, relying on online sources can get dangerous, even though they are a huge part of how we connect socially. The thesis points out that line where it stops being helpful and starts being risky. Some people might think its not that bad, but I am not totally sure. Anyway, this should highlight why we need better ways to handle it.

Introduction

In the modern digital age, the internet, and more specifically, social networks, have become the main source through which information diffuses rapidly[1]. Information in text form, whenever it appears on the internet, has the capability of becoming viral in a matter of minutes. Most people are dependent on the news they receive through social networks without bothering to check if it is true or false. They just read it without questioning it, thus contributing largely to the creation of false pieces of information[4][11].

The extent to which the COVID-19 pandemic brought to light the negative effects of misinformation can be seen in the fact that, during that period, a significant amount of information

was shared daily via social platforms, messages, and online news publications. Nevertheless, most of this information is either inaccurate or partial. In a hypothetical situation, only about 20% of shared information is authentic, while about 80% comprises misleading or inaccurate information[2]. As a result, there has been fear, confusion, and panic on the part of individuals due to increased misinformation generated by this period.

Despite the current surroundings, deceptive text information about public personalities, contentious topics, and trending issues continues to be widely circulated through social networking sites. Misleading statements, carefully designed text, and deceptive titles are used for malicious tasks such as reputation establishment, as well as deceiving society at large[3]. Owing to the speed of social networking sites, this kind of deceptive information gets widely disseminated, and society remains unaware of it as it gets shared without any confirmation[4].

In this scenario, the need has arisen for an individual to verify the validity and contents of the information before utilizing it. Regarding Computer Science, the machine learning techniques and natural language processing have been found to act as an efficient means to deal with texts[3][4]. The research study is dependent on the identification and classification of the misinformation based on texts, and is classified into real information, fake information, and incomplete information[9]. In this research study, some concrete steps would be taken to overcome the critical problem pertaining to the dissemination of misinformation and the gap that arises in the research study to generate a safe environment for information[10].

Literature Survey

Recently, the problem of misinformation detection has gained considerable attention because of the growing use of digital media and social networking sites[11]. In various research works, the need for managing the unbridled spread of incorrect and deceptive information has been underlined as having generated a serious international concern because of its implications and consequences for society and opinion formation. In the context of Public Health concerns and Climate Change problems, the problem of misinformation has been found to generate catastrophic implications[5]. Because of such considerations and implications for society and opinion formation, various researchers belonging to a variety of disciplines have made serious efforts towards the development of automated systems that can counter the effects of misinformation[3][4].

Many existing studies relate to data collection approaches where large volumes of textual data are gathered from news media websites, social media sites, and open data repositories. In these datasets, there are labeled samples of genuine and fake news[6]. The need for having a representative dataset in the design of a robust system capable of detecting fake news is highlighted in previous studies[12].

The majority of the literature focuses on the application of NLP processes for text processing and feature derivation. Techniques include tokenization, removal of stop-words, stemming or lemmatization of texts, and vectorization methods, such as Bag of Words and TF-IDF[7]. Traditional information retrievals have shown that proper weighing of the term helps in determining the relevance of the term to a document. Techniques have assisted in extracting relevant linguistic patterns from texts. This is important for proper text classifications[8].

Previous studies have proposed several machine learning models, such as Naive Bayes, Support Vector Machines, Logistic Regression, Random Forests, and deep learning-based models[3][4]. Studies have shown that traditional machine learning classifiers can be competitive in text-based misinformation detection tasks if they are combined with appropriate feature extraction techniques[3][8]. More advanced deep learning

architectures, such as convolutional neural networks and attention-based models, further improved performance by capturing semantic and contextual relationships[4][7]. However, one observes from cross-analyses across different studies that no single model has consistently outperformed others across all datasets, with performance often depending on the nature of the data and the features used[3][4].

Almost all the review papers consistently emphasize the fact that misinformation is a graver and worsening problem that needs serious attention[4][5]. The urgent need for detection mechanisms to be reliable, scalable, and automated to meet the increasing volume and speed of false information dissemination is stressed by global organizations and academic researchers alike[5][6]. While much progress has been attained in this regard, misinformation detection remains an open problem in terms of adaptability, explainability, and deployment in a real-world scenario[7].

Overall, the literature has set a very good foundation for the present research by providing an understanding of methods of data collection, strategies of preprocessing, techniques of feature extraction, and machine learning models[4][6][8]. The findings have inspired and supported conducting this research, further developing the previous work, and proposing a text-based machine learning approach that might contribute to overcoming the increasingly evolving problem of misinformation[3][5].

Proposed Methodology

This proposal will introduce a method of detecting misinformation through texts using machine learning and natural language processing. The method will be able to determine whether news is real or fake based solely on texts. The whole process will include data retrieval, preprocessing, development, and deployment.

1. Data Collection

Several publicly available textual news datasets have been gathered from trustworthy open sources, such as Kaggle datasets and research datasets hosted on GitHub. Even though the datasets are from different sources, the labels are all uniform

with two classes: real and fake news. Given that the core of the detection of misinformation is identifying fake information as opposed to genuine information, all the datasets have been merged.

Combining data from various sources helped in increasing the diversity and reducing the bias in the data, thus enhancing the generalization capacity of the proposed system.

```
df = pd.read_csv(r"C:\Research Paper\fp\project\data\dataset_combined.csv")
df.head()
```

	text	label
0	Did Miley Cyrus and Liam Hemsworth secretly ge...	fake
1	Paris Jackson & Cara Delevingne Enjoy Night Ou...	fake
2	Celebrities Join Tax March in Protest of Donal...	fake
3	Cindy Crawford's daughter Kaia Gerber wears a ...	fake
4	Full List of 2018 Oscar Nominations – Variety	fake

Fig.1 Data collection

2. Data Preprocessing

In order to process raw text data for machine learning models, a structured process was used in the preprocessing step. The overall process involved a series of tasks:

- Making text lowercase.
- Conversion of texts to tokens or words.
- Elimination of non-alphanumeric characters.
- Removing Stop Words using the stopwords list in NLTK.

	text	label	transformed_text
0	Did Miley Cyrus and Liam Hemsworth secretly ge...	fake	miley cyru liam hemsworth secretli get marri
1	Paris Jackson & Cara Delevingne Enjoy Night Ou...	fake	pari jackson cara delevingn enjoy night match ...
2	Celebrities Join Tax March in Protest of Donal...	fake	celebr join tax march protest donald trump
3	Cindy Crawford's daughter Kaia Gerber wears a ...	fake	cindi crawford daughter kaia gerber wear wig d...
4	Full List of 2018 Oscar Nominations – Variety	fake	full list 2018 oscar nomin varieti

Fig.2 Data Preprocessing

3. Feature Extraction

Since machine learning algorithms cannot process raw text directly, the preprocessed text was transformed into numerical form using the Term Frequency–Inverse Document Frequency (TF-IDF) technique. TF-IDF assigns importance weights to words based on their frequency within a document and throughout the corpus, to enable effective representation of the text information.

The TF-IDF matrix obtained was then utilized to train and test all machine learning models.

```
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer

# Create vectorizers
cv = CountVectorizer()
tfidf = TfidfVectorizer(max_features=3000)

X=tfidf.fit_transform(df['transformed_text']).toarray()
```

Fig.3. Feature Extraction

4. Model Training

The data is split into training and testing sets using an 80/20 split. Various machine learning classifiers are used in separate attempts to classify misinformation in text.

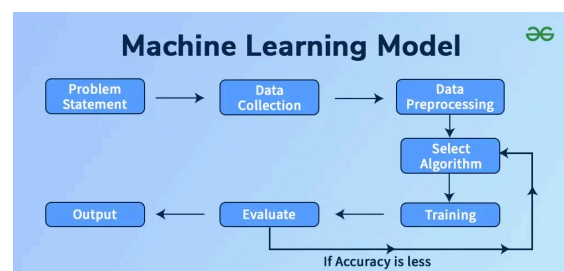


Fig.4 Model Training

1. Bernoulli Naive Bayes

It is a type of probabilities-only classifier, which deals with the binary features of a problem, specifying the presence or absence of words in a document, and can be used when the analysis of text involves important features like the presence or absence of a word in a document.

2. Multinomial Naive Bayes (Final Model)

It is a commonly employed technique in the classification of texts because it takes into account the frequency of the words. The algorithm is quite efficient in its operation when the features are TF-IDF. For its high efficiency and low computational operation cost, the final classifier was chosen as this model.

3. Complement Naive Bayes

This is an advanced level of Multinomial Naive Bayes and is applied when dealing with class imbalance problems. The model works best when it is given imbalanced samples of actual and fake news.

4. Logistic Regression

It is a linear classifier with a probability estimation problem, trying to measure how likely it is to see a particular vector among a set of possible vectors, given a specific class. It is suited to big datasets, especially those with text.

5. Linear Support Vector Classifier

Linear SVC is a margin-based classifier that separates classes by maximizing the margin between the boundaries of the classes. It performs very well on high-dimensional text data and applies to TF-IDF transformations.

6. Random Forest Classifier

Random Forest is a type of Ensemble Learning technique that incorporates several Decision Trees together with a goal of increasing the precision of predictive analysis and preventing overfitting at the same time. It discovers non-linear associations within data.

7. Stochastic Gradient Descent Classifier

SGD Classifier is an efficient model based on optimization and can handle large datasets effectively. It performs extremely well on linear models, SVM, and Logistic Regression when using stochastic gradient descent algorithms for training purposes.

8. Multi-Layer Perceptron Classifier

MLP is a feed-forward neural network, which is proficient in learning complex patterns from the input data. It represents the nonlinear associations between features and is a deep learning solution for text categorization. A common set of TF-IDF feature representations was used for training each of the models to ensure a balanced comparison.

5. Model Evaluation

The accuracy of the models was assessed on the test dataset with accuracy as the major criterion. Confusion matrices were employed to compare the classification accuracy of the models on the basis of their true positive, true negative, false positive, or false negative outputs.

All the above models were compared through a comparative analysis in a bid to establish the most effective and efficient model capable of classifying misinformation from the text.

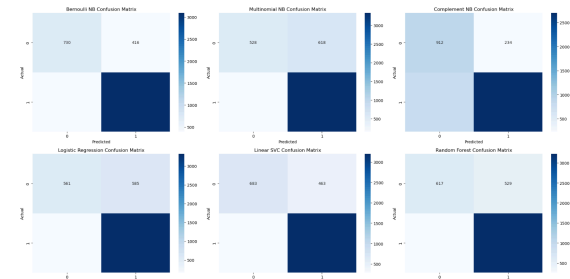


Fig.5 Model Evaluation

6. Final Model Selection

According to the result of experimental evaluation, the classification system was identified as Multinomial Naive Bayes. This system was found to produce high precision with lower complexity. Again, the system was identified by its name, that is, Multinomial Naive Bayes. This system is effective in classifying the texts because it can handle the sparse vectors that are generated by the TF-IDF transformation.

This is a very efficient approach with a simple design, fast speed for its trains relative to other models, and strong performance.

7. Model Deployment

The trained Multinomial Naive Bayes classifier and TF-IDF vectorizer were pickled, meaning made persistent, using the job library. During the deployment stage, the incoming text form of news entries would transform using the persistent vectorizer and then be classified by the trained classifier to detect news items as genuine and fraudulent.

This will ensure consistency between the training process and the evaluation process. Additionally, this will make it possible to utilize the system for real-world tasks.

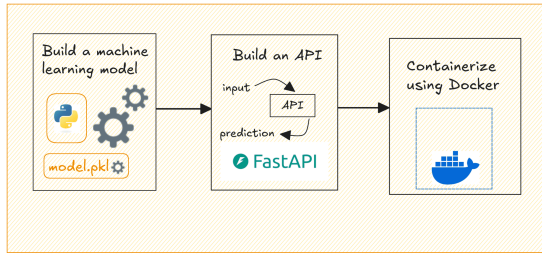


Fig.6 Model Deployment

8. Scope of the System

Its approach is purely concerned with the identification of misinformation through text, and it does not require any image, video, or multimedia files. Therefore, it is lightweight and scalable and can serve as a news verification tool, a social media surveillance tool, and a content management tool.

Results and Analysis

To assess the efficacy of the proposed text-based misinformation detection model, a variety of machine learning models were trained and tested on the same TF-IDF matrix. The performance of the models was measured using the measures of accuracy and precision. Accuracy and precision are more important in the context of misinformation detection since accurate detection of misinformation is essential.

Table 1:- Model comparison

Model Name	Accuracy	Precision	Recall	F1-Score	ROC-AUC
BernoulliNB	0.8249	0.8814	0.8867	0.884	0.8627
MultinomialNB	0.8327	0.8435	0.9549	0.8957	0.8637
LogisticRegression	0.8364	0.8492	0.9515	0.8975	0.8664
LinearSVC	0.8375	0.8735	0.9168	0.8947	0.859
RandomForestClassifier	0.8286	0.859	0.924	0.8903	0.8366
ComplementNB	0.7788	0.9201	0.7732	0.8403	0.8637
SGD_Classifier	0.839	0.8521	0.9513	0.8989	0.8585
MLP_Classifier	0.8155	0.8802	0.8739	0.877	0.8467

Presents the comparative performance of all models implemented in terms of accuracy score and precision score.

Comparative Performance of Models

Based on the experimental analysis, it has been noticed that the linear classifiers, which include Logistic Regression, Linear SVC, and SGD Classifier, performed best. This is mainly because they have high accuracy scores with a good combination of precision and recall, which reveals their excellence in dealing with high-dimensional

sparse text feature extraction using TF-IDF. Additionally, it has been noticed that Logistic Regression and Linear SVM performed evenly well, whereas the SGD Classifier performed slightly better on accuracy metrics, which again validates that linear classification boundary works best for text classification tasks.

Comparing various probabilistic classifiers, Multinomial Naive Bayes (MNB) was found to be the best method, and it resulted in an accuracy of 0.8327 and an F1-score of 0.8957. Multinomial Naive Bayes was considerably better than Bernoulli NB and Complement NB in terms of recall value (0.9549), indicating that it is more efficient in spotting the relevant text examples. It is applicable in situations where term frequencies are taken into account.

Moreover, MultinomialNB is computationally efficient and works well even with high-dimensional data and very fewer samples of training data. This makes it a good baseline algorithm for large-scale datasets related to text classification.

Although Random Forest and MLPClassifier were included to test ensemble-based and neural network approaches, results did not beat those obtained by simpler linear and probabilistic models. Therefore, in this type of problem with TF-IDF-based textual data, complex models do not ensure better results, besides being overfitted or computationally expensive.

These results clearly show that linear models and Multinomial Naive Bayes achieve the best trade-off between performance, efficiency, and interpretability; thus, these two are the best choices for this text classification task.

Final Model Selection

Considering that some of the models had slightly higher accuracy, Multinomial Naive Bayes was selected as the final classification model in this research. This was based on the following reasons:

- Strong balance between precision and accuracy.
- Handling sparse textual data based on TF-IDF with efficiency.

- Less time is taken to train the model and generate predictions.
- Lower computational complexity compared to more complex models.
- Good performance on several datasets.

Precision plays a great role in misinformation detection, as it decreases the number of instances where real news could be marked as fake. Multinomial Naive Bayes has given a competitive precision with very dependable accuracy, making it a good candidate for practical deployment.

The experimental results confirm the effectiveness of the proposed textual misinformation detection framework and validate the selection of multinomial Naive Bayes as the final model.

Conclusion

As deduced in the above study, one of the most important issues in the new world of digital technology is misinformation identification, as people rely on the internet and social media platforms without filtering the information, or even checked for that matter. For such an issue, a text-based ML model has been proposed in the background of news classification as genuine and non-genuine news that is not.

Several machine learning models were trained on the preprocessed data containing text using the TF-IDF technique for feature extraction. A total of eight machine learning models were tested for their skill in matters of accuracy and precision. Out of all the models that were trained on the dataset, the best model based on its efficiency is observed to be the Multinomial Naive Bayes model. Considering the capability of the Multinomial Naive Bayes model in dealing with text data efficiently using word frequency, the same model is employed for the proposed system.

It is evident from the experimental result that the detection of trustworthy information is possible with only text information and that there is no necessity for addenda information such as images, videos, and metadata. The proposed system is light and is an efficient one too. It proves that probabilistic and linear learning machines with quality results are possible by mixing appropriate preprocessing and extraction techniques.

Overall, this research is an additional effort in the bridge created among the studies concerning the mis- detection of misinformation. The developed model can be readily incorporated into various platforms, like the news-verification platform and social platform for monitoring, which help in decision-making through proper information.

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